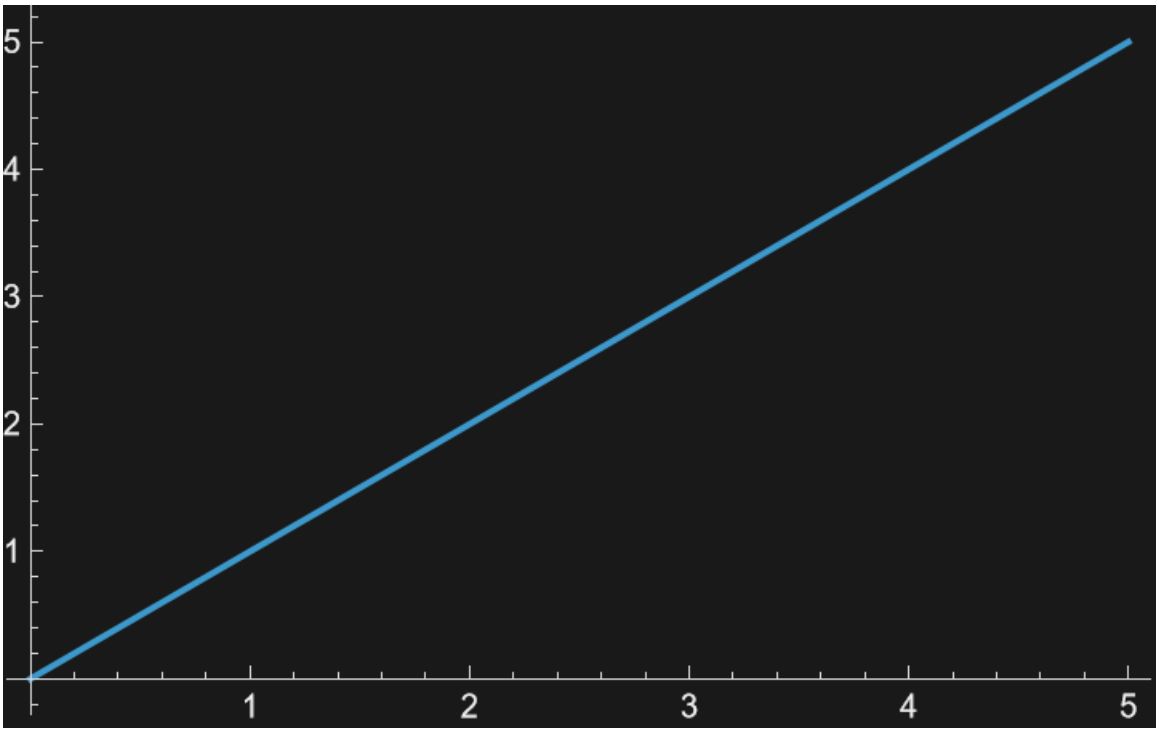
Lab 5 Solutions

In[1]:=

```
(* Problem 1 - Plot y = x, with  [0,5]*)

Plot[x, {x, 0, 5}]
```



In[2]:=

```
(* Problem 2 - Find 1st and Second Derivatives*)

(*Create the function f(t)*)
f[t] := α*Log[a*t]*Cos[ω*t]*Sin[ω*t]

(*Call First Derivative*)
D[f[t], t]

(*Call Second Derivative*)
D[f[t], {t, 2}]
```

$$\alpha\,\omega\,\text{Cos}[t\,\omega]^2\,\text{Log}[a\,t] + \frac{\alpha\,\text{Cos}[t\,\omega]\times\text{Sin}[t\,\omega]}{t} - \alpha\,\omega\,\text{Log}[a\,t]\,\text{Sin}[t\,\omega]^2$$

$$-\alpha\,\omega^2\,\text{Cos}[t\,\omega]\times\text{Log}[a\,t]\times\text{Sin}[t\,\omega] - 2\,\alpha\,\omega\,\text{Sin}[t\,\omega]\left(\omega\,\text{Cos}[t\,\omega]\times\text{Log}[a\,t] + \frac{\text{Sin}[t\,\omega]}{t}\right) + \alpha\,\text{Cos}[t\,\omega]\left(\frac{2\,\omega\,\text{Cos}[t\,\omega]}{t} - \frac{\text{Sin}[t\,\omega]}{t^2} - \omega^2\,\text{Log}[a\,t]\times\text{Sin}[t\,\omega]\right)$$

In[3]:=

```
(*Problem 3 - Find Derivative to Arbitrary Function w(x)*)

(*Create w(x)*)
Clear[f]
```

```
w[x] := f[x]*g[x]*h[x]

(*Call 1st Derivative*)
D[w[x], x]
```

$$g[x] \times h[x]\, f'[x] + f[x] \times h[x]\, g'[x] + f[x] \times g[x]\, h'[x]$$

In[4]:=

```
(*Problem 4 - Solve Equation of Motion*)

(*Initialize Omega, l, g*)
ClearAll["Global`*"]
ω0 = Sqrt[(g/l)]
l = 10
g = 980

(*Call NDSolve[]*)
sol = NDSolve[
  {φ''[t] == -ω0^2 Sin[φ[t]],
   φ'[0] == 0,
   φ[0] == 30 Degree},
  φ, {t, 0, 2}]

Plot[Evaluate[φ[t] /. sol], {t, 0, 2},
  AxesLabel -> {"t (s)", "φ (rad)"},
  PlotLabel -> "Pendulum, φ0 = 30°"]
```

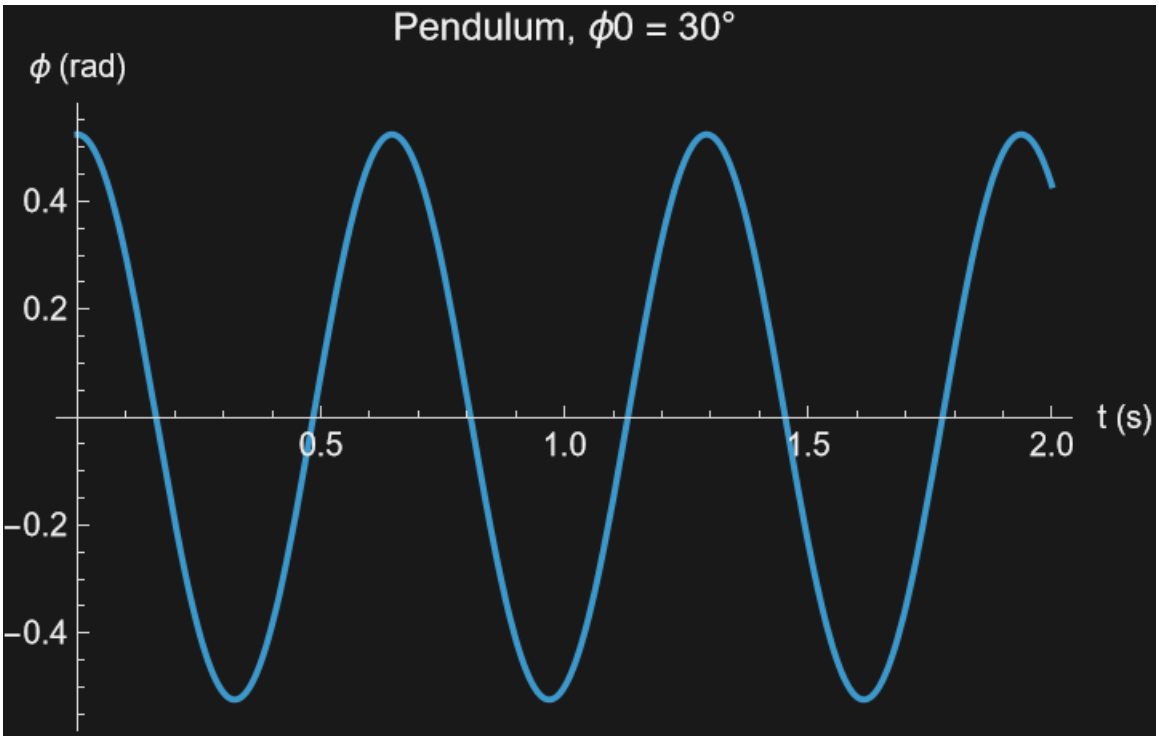
$$7\sqrt{2}$$

10

980

$$\{\{\phi \to \text{InterpolatingFunction}[\langle\langle 1 \rangle\rangle]\}\}$$

Domain: {{0., 2.}}
Output: scalar



In[5]:=

```
(*Problem 5 - 3D Torsion Pendulum Problem*)
Graphics3D[{Gray, Cylinder[{{0, 0, 0}, {0, 0, 1}}, 2], Red, Cylinder[{{0, 0, 0}, {0, 0, 5}},
0.2]}]
```

